09. Business Interpretation, Results & Limitations

Summarise what the project demonstrates and what must be fixed before real use.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

What the project demonstrates

The project demonstrates an end-to-end tabular classification workflow: data inspection, label encoding, exploratory plots, train/validation/test splitting, Random Forest and SVM experiments, class balancing with SMOTE, hyperparameter search and serialized model artifacts.

Main findings

- The original data has 837 rows and four target classes with extreme imbalance.
- Stunting dominates the sample (83.2%); Wasting has four examples (0.5%).
- The simple SVM branch reports 84.13% test accuracy, but minority-class recall is poor and Wasting is missing from the test partition.
- The training-only SMOTE branch is methodologically preferable to the whole-dataset SMOTE branch, but its reported RF test accuracy is only 64%.
- The dense neural network uses a binary output/loss for a four-class target and should not be considered a valid multiclass implementation.

Critical repository issues

- Global SMOTE before splitting in Oversampling on the whole dataset.ipynb causes data leakage and invalidates downstream evaluation.
- Splits do not use stratify=y; the test set has zero Wasting observations.
- One neural-network branch uses one sigmoid output and binary_crossentropy for four labels.
- Some notebooks drop the three income columns more than once in successive cells, which can trigger KeyError depending on execution order.
- The saved Xtrain.csv and resampled_data.csv are outputs of the globally oversampled workflow and should not be treated as clean raw training data.
- The FastAPI app has a hard-coded local filesystem path to a model file; model artifact paths and feature schemas are not portable.
- The app repeats a Sex conversion branch; it is redundant, and input validation does not check physiologically plausible ranges.

Ethical and practical limitations

Nutrition-related labels are sensitive and may affect access to care or public services. Errors can disproportionately affect underrepresented groups. The dataset is small, provenance and units are unclear, and the labels may encode local measurement or policy assumptions. Do not use predictions to deny services or make automated clinical decisions.

Recommended next steps

- Confirm label definitions, measurement units and population with a qualified domain expert.
- Collect more examples, especially Wasting and Underweight, across age/sex/population strata.
- Create a reproducible stratified or group-aware split before any resampling.
- Compare class weights, random oversampling and SMOTE inside cross-validation pipelines.
- Use macro-F1, per-class recall, confusion matrices, calibration and uncertainty intervals.

- Use clinically meaningful age/sex-specific growth references and validate against expert assessment.
- Perform subgroup error analysis and independent external validation before any operational use.

Interview conclusion

> “The most important learning from this project was that imbalance handling and evaluation design matter more than a high headline accuracy. The baseline models learned the dominant Stunting class, but minority-class performance remained weak. I would not claim this as a reliable screening model without more minority data, leakage-free validation and domain validation.”